OLLSCOIL NA GAILLIMHE
UNIVERSITY OF GALWAY

Semester 2 Examinations 2024/2025

Course Instance Code(s)	4BCT, 3BP, 4BP, 3BLE, 4BLE, 1OA, 1EM
Exam(s)	B.Sc. (Computer Science & Information Technology), B.E.(Electronic and Computer Engineering), B.E. (Electrical & Electronic Engineering)
Module Code(s)	CT420
Module(s)	Real-Time Systems
Bonded	No [☑] Yes [ ]
Paper No.	1
External Examiner(s)	Dr. Ramona Trestian
Internal Examiner(s)	Dr. Enda Howley *Dr. Jawad Manzoor *Dr. Michael Schukat

Instructions: This exam has two sections:
In section A, answer any *two* questions between Question 1, 2, and 3.
In section B, answer any *two* questions between Question 4, 5 and 6.

Duration	**2 hours**
No. of Pages	**5**
Discipline(s)	Computer Science
Course Co-ordinator(s)	Dr. Colm O'Riordan

Requirements:

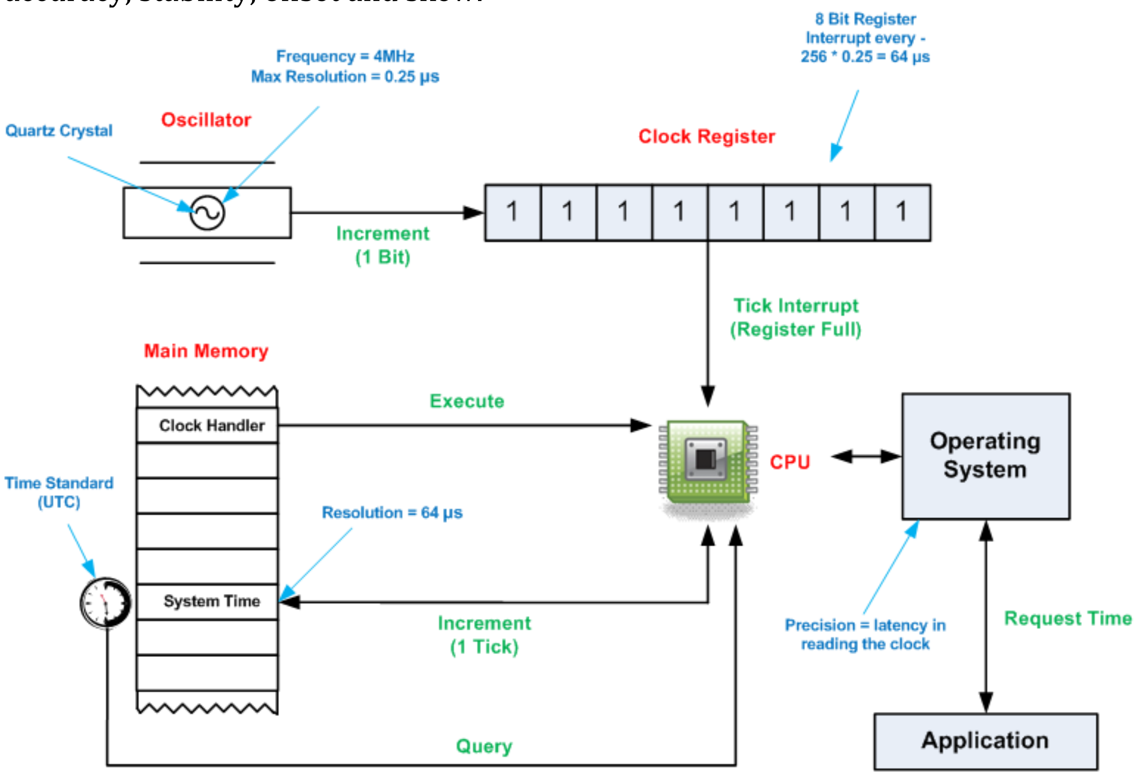
Release in Exam Venue	No[ ]	Yes [☑]
MCQ Answer sheet	No [☑]	Yes [ ]
Handout	No [☑]	Yes [ ]
Formulae & Tables*	No [☑]	Yes [ ]
Cambridge Tables 2nd Edition**	No [☑]	Yes [ ]
Graph Paper*** A4 Graph Paper 1mm 0.1cm Squared (Standard)	No [☑]	Yes [ ]
Other Materials	No [☑]	Yes [ ]
Graphic material in colour	No [ ]	Yes [☑]

End of requirements.

Section A
Answer any two questions out of three in Section A

Question 1 (15 Marks)

(a) Using the diagram below, discuss how **crystal accuracy** and **crystal stability** impact on the performance of computer clocks, in terms of **clock offset** and **clock skew**. In your answer, provide definitions for the terms accuracy, stability, offset and skew.



[5 marks]

(b) Using (pseudo) C-code, design a **Clock Handler** (i.e., clock interrupt service routine) that keeps track of time and allows for clock skew compensation. Fully explain your code. For your answer, you can assume that the following POSIX data structure is used:

```
struct timespec {
        time_t tv_sec;
        time_t tv_nsec;
}
```

[5 marks]

(c) An alternative way to correct computer clocks is by using an internal clock synchronisation algorithm like the **Berkeley Algorithm**. Using an example, show how this method works, and how local clocks that are completely out of sync are dealt with.

[5 marks]

PTO

Question 2 (15 Marks)

(a) In relation to logical clocks, what is meant by the term **causality**? Using an example show why **Lamport's logical clocks** do not capture causality, while **vector clocks** do capture causality.

[5 marks]

(b) Briefly discuss the following terms, their inner workings and benefits for accurate PTP time synchronisation:

a. Sync and Follow_Up messages [1 mark]
b. Transparent clocks [1 mark]
c. Hardware timestamping [1 mark]
d. P2P versus E2E delay calculations [2 marks]

[5 marks]

(c) Do an RM schedulability analysis on the set of 4 tasks shown below, focussing on W_4 and its associated checkpoints. Subsequently, determine the RM task schedule over the interval [0, 400]:

i	e_i	p_i
1	20	100
2	20	180
3	80	240
4	100	400

[5 marks]

PTO

Question 3 (15 Marks)

(a) Using (pseudo) C-code develop a cyclic executive scheduler for the set of tasks below. Your code must be able to detect task overruns. Fully explain your answer.

Task	Period p [ms]	Exec Time [ms]
A	25	10
B	25	8
C	50	5
D	50	4
E	100	2

[5 marks]

(b) What are the benefits of the Priority Inheritance Protocol and what problem does it solve? Provide an example including diagrams to support your answer.

[5 marks]

(c) Memory locking is an important feature of POSIX.4. Use the 7-state process model in conjunction with an example written in C to outline how memory locking works. Use pseudo code if you can't recall the exact POSIX API calls and explain what would happen if memory locking was not used.

[5 marks]

PTO

Section B
Answer any two questions out three in Section B

Question 4 (15 Marks)

a) Explain the primary motivations for developing the HTTP/2 protocol, focusing on the limitations of HTTP/1 that prompted the transition. Discuss *three* new features introduced in HTTP/2.

[7 Marks]

b) Explain how latency affects the performance of soft real-time systems. Using a diagram, compare the connection establishment process of QUIC and TCP. Discuss how QUIC reduces latency in communication.

[8 Marks]

Question 5 (15 Marks)

a) Compare and contrast Wireshark and qLog for network traffic analysis.

[5 Marks]

b) Evaluate the benefits of using qViz for debugging performance issues with stream multiplexing and congestion control.

[5 Marks]

c) Discuss how QUIC's congestion control approach helps reduce latency and improve overall performance.

[5 Marks]

Question 6 (15 Marks)

a) Explain the importance of Quality of Service (QoS) in soft real-time systems. Differentiate between Perceived QoS and Intrinsic QoS.

[5 Marks]

b) Explain the role of a jitter buffer in a VoIP system and why it is critical for ensuring smooth audio playback. Compare and contrast fixed and adaptive jitter buffer strategies. Highlight the advantages and disadvantages of each approach.

[6 Marks]

c) Define response delay in the context of cloud gaming and explain its key components, including processing delay, network delay, and playout delay.

[4 Marks]

END OF EXAM